Venkata Lakshmi Parimala Pasupuleti

Full-Stack AI Engineer · Python · TypeScript · React · LLM Integrations · RAG Pipelines · Eval Frameworks · Distributed Systems · AI-Native Internal Tooling

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PROFESSIONAL SUMMARY

Full-stack software engineer with 8+ years shipping production systems at scale (1M+ records, 100K+ MAU), now specializing in **AI-native internal tooling** — Python REST API backends, React frontends, and **LLM integrations** that make engineering workflows faster and more reliable. Built **AgentProbe**: a multi-agent AI evaluation framework with a 15-endpoint FastAPI backend, React observability dashboard, **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance), **model monitoring** via per-run regression drift tracking, and CI/CD gates blocking failing agent runs before production — running 100% locally at zero cost. Built **Leasel**: an agentic **RAG pipeline** over 1,240 legal PDFs achieving 93.6% precision using **prompt engineering**, ChromaDB **semantic search**, and a deterministic rules engine with LLM-as-judge scoring. 8 years of distributed-systems engineering at Entrata (microservices, **asynchronous job execution** via RabbitMQ, AWS, 99.5% uptime). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

PROJECTS

AgentProbe: Multi-Agent AI Evaluation & Monitoring Framework

LLM APIS · EVAL ENGINE · MLOPS

Python · LangGraph · FastAPI · React · TypeScript · ChromaDB · OpenAI API · Anthropic API · Groq · Ollama · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Deployed a **15-endpoint FastAPI REST API backend** + React TypeScript observability dashboard — full-stack AI-native internal tooling running 100% locally at zero cost; **asynchronous job execution** across concurrent multi-agent test suites with per-run latency and regression drift tracking.
- Built a **5-metric LLM-as-judge evaluation framework** (factual accuracy, hallucination, retrieval, completeness, compliance) with **model monitoring**, **model version control** via CI/CD regression gates, and HITL feedback loops — blocks failing agent runs before production on every commit; implements **MLOps** principles end-to-end: deployment pipelines, versioning, eval, monitoring; compliance metric explicitly surfaces **bias and fairness** in AI outputs, flagging inconsistent or discriminatory model behavior before it reaches users.
- Engineered a **multi-provider LLM integration layer** (OpenAI, Anthropic, Groq, Ollama) with dynamic routing via **prompt engineering** strategies and 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) — inter-agent state handoffs, tool calling, and a **semantic search** RAG layer over ChromaDB.

Leasel: Agentic RAG Pipeline for Legal Document Analysis

RAG · LLM-AS-JUDGE

Python · LangGraph · ChromaDB · pgvector · Prompt Engineering · Deterministic Rules Engine · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** on 1,240 complex legal PDFs — agentic LangGraph **retrieval-augmented generation** pipeline: extraction, chunking, embedding, **semantic search** (ChromaDB + pgvector), deterministic rules engine for fully auditable high-stakes outputs; **prompt engineering** strategies per compliance rule type.
- Shipped a **pass@k evaluation framework** (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge scoring across accuracy and reasoning quality — CI/CD regression gates; full **MLOps** lifecycle: versioning, quality gates, production validation.

AI on FHIR: NLP-Driven Medical Records Query Platform

HEALTHCARE · FULL-STACK

Python · Flask · React · TypeScript · FHIR APIs · LLMs · BERT · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in clinical retrieval time — end-to-end **full-stack AI tooling**: Python/Flask REST API backend + React TypeScript frontend translating natural-language queries (EN/ES) to FHIR API calls via **LLM integrations**, BERT embeddings, and semantic search; HIPAA-compliant RBAC across structured medical records.

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic Systems · Camera Imaging Pipelines · LLM Integrations · RAG · Prompt Engineering

- Built an agentic mission-planning system as a LangGraph stateful graph with **asynchronous job orchestration** across pipeline stages; RAG layer over FAA regulatory corpus with cited sources; deterministic validation guardrails blocking non-compliant missions before export.
- Engineered **camera image quality pipelines** for aerial survey operations — parsed natural-language requests into structured flight plans using quantitative **photogrammetry math** (ground sampling distance, **camera footprint**, overlap-driven line spacing); computed serpentine survey grids ensuring consistent **image quality** and coverage across marketing, real estate, and commercial survey domains — direct experience with **image data and metadata pipelines** in a production engineering context.

George Mason University

Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Statistical Modeling · Anomaly Detection

- Built end-to-end ML pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; automated reporting with Python + Quarto driving **40% improvement** in stakeholder comprehension.

Entrata (PropTech SaaS — 100K+ MAU, 1M+ records)

Jan 2017 – Aug 2024

Senior Software Engineer → Associate Engineer — Full-Stack · Distributed Systems · Production AI · AWS · Cross-functional Delivery

- Owned **20+ production full-stack modules** end-to-end (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints — architecture, **REST API design**, implementation, rollout — **1M+ records, 100K+ MAU, 99.5% uptime**; 300+ peer code reviews; mentored 5+ engineers in maintainability and correctness practices.
- Led **distributed-systems initiatives**: Docker microservices rollout (50% reliability gain, 30% deployment time reduction); AWS S3/RDS/Glue migration (100% PHI compliance, zero incidents); **asynchronous job execution** via RabbitMQ and Python/Bash pipelines eliminating 80% of manual operational workload.
- Shipped **Entrata's first production AI pipeline** on 1.6M real financial records (PyTorch, OpenAI API) — automated data processing workflows with quality gates, output validation, and HITL review; **cross-functional collaboration** across compliance, operations, and product stakeholders throughout design, build, and rollout.
- **Optimized 200+ SQL queries** (30% performance gain) via execution plan tuning, indexing, **Redis caching**, and database partitioning; led database architecture decisions; built internal debugging tools reducing debugging time 40%; **100% PHI compliance** via Libsodium encryption across entire dataset.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

ML systems, statistical modeling, data engineering, AI pipelines, distributed systems, evaluation methods

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Languages & Full-Stack: Python, JavaScript, TypeScript, SQL, PHP, R; FastAPI, Flask, Django, REST API design, React, HTML/CSS; asynchronous job execution (RabbitMQ, Python async), webhooks, event-driven distributed systems

AI & LLM: LLM integrations (OpenAI API, Anthropic API, Groq, Ollama), LangGraph, LangChain, RAG pipelines, semantic search, prompt engineering, vector databases (ChromaDB, pgvector), LLM-as-judge, evaluation frameworks, model monitoring, MLOps (CI/CD eval gates, versioning, regression tracking), bias and fairness in AI, HITL feedback loops, NLP, BERT, PyTorch, Hugging Face

Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, microservices, CI/CD (GitHub Actions), PostgreSQL, MySQL, Redis (caching, session management), test-driven development, HIPAA/PHI compliance, Git, Linux, Agile/Scrum